



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-04 Targeted Remediation Cards

Repair only the failed competency, then prove it on a changed micro-case.

Artifact	C3-04-REPAIR
Status	2026 - LEARNER EDITION
Phase	C3-04 Storage, File Formats & Lakehouse 13 lessons 34-46 focused hours
Prerequisite	C3-03 Gate PASS; Bridge warehousing/lakehouse mental models assumed, implementation depth taught here.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: mental picture -> tiny example -> guided real file/table workflow -> break/fix -> independent changed task -> validation -> production reasoning -> evidence. A file extension or successful write is never sufficient proof of correct table state.



Repair rules

Do not restart the phase for one narrow weakness. Save the failed evidence, use the matching card, close review and solve a changed micro-case. Completing repair never auto-assigns Gate B/C.

R-LAYOUT - Format/partition/layout

Choose a different workload, estimate predicates/cardinality/file count and defend a fresh format + partition decision.

PASS: changed evidence is correct, reproducible and explainable with review closed. Do not reuse the original Gate case.

R-PARQUET - Parquet metadata

Write a fresh small Parquet file and save real schema/row-group/projection evidence plus source parity.

PASS: changed evidence is correct, reproducible and explainable with review closed. Do not reuse the original Gate case.

R-TXN - Delta committed state

Create a fresh Delta table, reopen it, record version/schema/rows and distinguish data files from `_delta_log`.

PASS: changed evidence is correct, reproducible and explainable with review closed. Do not reuse the original Gate case.

R-SCHEMA - Schema evolution

Classify two new changes - one additive and one breaking - then implement only the allowed one and validate downstream shape.

PASS: changed evidence is correct, reproducible and explainable with review closed. Do not reuse the original Gate case.

R-MERGE - Grain-safe upsert

Use a new key/domain. Prove source uniqueness, MERGE/update+insert, target uniqueness, metric reconciliation and rerun convergence.

PASS: changed evidence is correct, reproducible and explainable with review closed. Do not reuse the original Gate case.

R-HISTORY - Historical read/retention

Create another table change, compare prior/current versions, then state the retention condition required to keep that read reproducible.

PASS: changed evidence is correct, reproducible and explainable with review closed. Do not reuse the original Gate case.

R-OPS - Compaction/retention

Diagnose a different file-health pattern; propose compaction, dry-run cleanup and recovery-safe retention with before/after controls.

PASS: changed evidence is correct, reproducible and explainable with review closed. Do not reuse the original Gate case.

R-ARCH - Catalog/layers/open table format

Design a second domain contract and conditional Delta-vs-Iceberg decision with explicit engines/catalog/operations assumptions.

PASS: changed evidence is correct, reproducible and explainable with review closed. Do not reuse the original Gate case.